

Stony Brook University
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ASSIGNMENT 1

SOLUTIONS

INSTRUCTIONS: Show all your work to get full points. This homework is meant to work out your knowledge from not only the probability review but also what you learned from ECO320: Mathematical Statistics. This is a good time to dig out your old notes and review some of the concepts.

1. Just prior to jury selection for O.J. Simpson's murder trial in 1995, a poll found that about 20% of the adult population believed Simpson was innocent (after much of the physical evidence in the case had been revealed to the public). *Ignore the fact that this 20% is an estimate based on a subsample from the population; for illustration, take it as the true percentage of people who thought Simpson was innocent prior to jury selection.* Assume that the 12 jurors were selected randomly and independently from the population (although this turned out not to be true).

- Find the probability that the jury had at least one member who believed in Simpson's innocence prior to jury selection. *Hint: Define the Binomial(12, .20) random variable X to be the number of jurors believing in Simpson's innocence.*
- Find the probability that the jury had at least two members who believed in Simpson's innocence *Hint: $\Pr(X \geq 2) = 1 - \Pr(X \leq 1)$ and $\Pr(X \leq 1) = \Pr(X = 0) + \Pr(X = 1)$.*

Solution

Let $X = \text{Bin}(12, 0.20)$ denote the number of jurors that believe in O.J.'s innocence. Since the opinion of each juror is identical and independent from one another, then this is indeed a binomial experiment.

- We need $\Pr(X \geq 1) = 1 - \Pr(X < 0) = 1 - \Pr(X = 0) = 1 - \binom{12}{0}(0.2)^0(0.8)^{12} = 0.9313$.
- Using the hint, $\Pr(X \geq 2) = 1 - \Pr(X < 2) = 1 - \Pr(X \leq 1) = 1 - [\Pr(X = 0) + \Pr(X = 1)]$. Thus, we only need to calculate $\Pr(X = 0)$ and $\Pr(X = 1)$.

$$\Pr(X = 0) = \binom{12}{0}(0.2)^0(0.8)^{12} = 0.06872,$$

$$\Pr(X = 1) = \binom{12}{1}(0.2)^1(0.8)^{11} = 0.20615.$$

Thus $\Pr(X \geq 2) = 0.72513$.

2. Suppose that a college student is taking three courses: a two-credit course, a three-credit course, and a four-credit course. The expected grade in the two-credit course is 3.5, while the expected grade in the three and four-credit course is 3.0. What is the expected overall grade point average for the semester? Remember that each course grade is weighted by its share of the total number of units.

Solution

The student is taking 9 credits in total. Therefore, the associated weights for the courses are, respectively: $\frac{2}{9}$, $\frac{1}{3}$ and $\frac{4}{9}$. Therefore, the expected GPA is $3.5 \times \frac{2}{9} + 3.0 \times \frac{1}{3} + 3.0 \times \frac{4}{9} = 3.11$

3. Much is made of the fact that certain mutual funds outperform the market year after year (that is, the return from holding shares in the mutual fund is higher than the return from holding a portfolio such as the S&P 500). For concreteness, consider a 10-year period and let the population be the 4,170 mutual funds reported in the Wall Street Journal on January 1, 1995. By saying that the performance relative to the market is random, we mean that each fund has a 50–50 chance of outperforming the market in any year and that the performance is independent from year to year.

- If performance relative to the market is truly random, what is the probability that any particular fund outperforms the market in all 10 years? *Hint: For any fund, performance across years is random, i.e. independent. In this question, you are being asked about a joint probability, which you can simplify with the help of independence.*
- Of the 4,170 mutual funds, what is the expected number of funds that will outperform the market in all 10 years? *Hint: For a binomial distribution $\text{Bin}(N, p)$, the expected value is given by Np .*
- Find the probability that *at least* one fund out of 4,170 funds outperforms the market in all 10 years. What do you make of your answer? *Hint: Let A be the event that at least one fund outperforms the market in all 10 years. Let B be its complement event, i.e. B is the event in which none of the funds outperforms the market in all ten years. The probability of B is easier to compute. Then $\Pr(A) = 1 - \Pr(B)$.*
- Use R to find the probability that at least five funds outperform the market in all 10 years. *Hint: You can use the commands `dbinom` or `pbinom` to perform this calculation.*

Solution

- Let $X_{i,t}$ denote the binary variable equal to one if fund i outperforms the market in year t . The assumptions of the problem put $\Pr(X_{i,t} = 1) = 0.5$, i.e. a 50–50 chance of outperforming the market for each fund in each year. The probability that fund i outperforms the market in all 10 years is given by

$$\Pr(X_{i,1} = X_{i,2} = \cdots = X_{i,10} = 1), \quad (1)$$

which is a joint probability over the 10 random variables $(X_{i,t})_{t=1,2,\dots,10}$. We are also assuming that performance across years is independent (as is told in the hint) so we can simplify equation (1) as

$$\Pr(X_{i,1} = 1) \times \Pr(X_{i,2} = 1) \times \cdots \times \Pr(X_{i,10} = 1) = \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}. \quad (2)$$

In fact, we can define a Bernoulli random variable X_i such that $X_i = 1$ if and only if fund i outperformed the market in all 10 years. then $\Pr(X_i = 1) = 1/1024$.

- Define the Binomial random variable X as $X = \sum_{i=1}^{4170} X_i$, i.e. X is *counting* the number of *successes* of 4170 independent Bernoulli trials, where each Bernoulli trial is given by the outcome of the random variable X_i defined in the previous item. Thus, X is counting the number of funds that in the 10 year period outperformed the market. Notice that $X \sim \text{Bin}(4170, 1/1024)$. Using the hint, we know that the expected value of X , i.e. the expected number of funds that outperform the market in the 10 year period, is $4170 \times (1/1024) = 4.07 \approx 4$.
- Let A denote the event in which $X \geq 1$, i.e. the event in which at least one fund outperforms the market in all 10 years. We seek to calculate $\Pr(A)$. We can calculate this probability using the notion of the

complement probability. Define B as the event in which $X < 1$, since X is a discrete random variable, this event can also be expressed as $X = 0$, i.e. this is the event in which *none of the funds* outperform the market in all 10 years. The probability of B is easier to calculate: $\Pr(B) = \Pr(X = 0) = \Pr(X_1 = X_2 = \dots = X_{4170} = 0) = \Pr(X_1 = 1) \Pr(X_2 = 1) \dots \Pr(X_{4170} = 1) = (1 - 1/1024)^{4170} \approx 0.0170$. Thus $\Pr(A) = 1 - \Pr(B) \approx 0.983$. This means that if performance relative to the market is random and independent across funds, it is almost certain that at least one of the funds will outperform the market in all 10 years.

- (d) We are looking for the probability that $X \geq 5$. Thus $\Pr(X \geq 5) = 1 - \Pr(X \leq 4)$. $\Pr(X \leq 4) = \Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2) + \Pr(X = 3) + \Pr(X = 4)$. In R you can use the command `dbinom(x,size,prob)` which gives the probability mass function of a Binomial random variable when x is the number of successes, `size` is the number of trials and `prob` is the probability of success:

```
> dbinom(0,4170,1/1024)+
+ dbinom(1,4170,1/1024)+
+ dbinom(2,4170,1/1024)+
+ dbinom(3,4170,1/1024)+
+ dbinom(4,4170,1/1024)
[1] 0.6147148
```

A more efficient way to perform this calculation:

```
> sum(dbinom(0:4,4170,1/1024))
[1] 0.6147148
```

Alternatively, we can use the command `pbinom(x,size,prob)` which gives the cumulative distribution function of a Binomial random variable:

```
> pbinom(4,4170,1/1024)
[1] 0.6147148
```

In any way you decide to do this, we have that $\Pr(X \geq 5) = 1 - \Pr(X \leq 4) = 1 - 0.6147148 = 0.3852852$. Thus, there is a non-trivial probability that at least 5 funds will outperform the market in all 10 years.

4. For a randomly selected county in the United States, let X represent the proportion of adults over age 65 who are employed, or the elderly employment rate. Then X is restricted to a value between zero and one. Suppose that the cumulative distribution function for X is given by $F(x) = 3x^2 - 2x^3$ for $0 \leq x \leq 1$. Find the probability that the elderly employment rate is at least 0.6 (60%).

Solution

We need to find $\Pr(X \geq 0.6)$. Thus

$$\Pr(X \geq 0.6) = 1 - \Pr(X \leq 0.6) = 1 - F(0.6) = 1 - (3(0.6)^2 - 2(0.6)^3) = 0.352.$$

5. This problem will require a little bit of calculus. Let X denote the prison sentence, in years, for people convicted of auto theft in a particular state in the United States. Suppose that the pdf of X is given by

$$f(x) = (1/9)x^2, \quad 0 < x < 3.$$

Find the expected prison sentence. *Hint: recall that the expected value of a continuous random variable is computed as $E[X] = \int x f(x) dx$.*

Solution

$$E[X] = \int_0^3 x \frac{1}{9} x^3 dx = \int_0^3 \frac{1}{9} x^4 dx = \left[\frac{1}{9} \frac{x^5}{5} \right]_0^3 = \left[\frac{x^5}{45} \right]_0^3 = \frac{3^5}{45} = \frac{243}{45} = \frac{9}{5}.$$

6. Suppose that a large university, college grade point average, *GPA* and the *SAT* score are related by the conditional expectation $E[GPA|SAT] = 0.70 + 0.002SAT$.

- Find the expected *GPA* when $SAT = 800$. Find $E[GPA|SAT = 1400]$. Comment on the difference.
- If the average *SAT* in the university is 1,100, what is the average *GPA*? *Hint: use the law of iterated expectations.*
- If a student's *SAT* score is 1,100, does this mean he or she will have the *GPA* found in part (b) of this question? Explain.

Solution

- $E[GPA|SAT = 800] = 0.70 + 0.002(800) = 2.3$. On the other hand $E[GPA|SAT = 1400] = 0.70 + 0.002(1400) = 3.5$. We can see that a student with a higher *SAT* score is predicted to have on average a higher *GPA*.

- We have that

$$\begin{aligned} E[GPA] &= E[E[GPA|SAT]] && \text{by the Law of Iterated Expectations,} \\ &= E[0.70 + 0.002SAT] && \text{by the definition of } E[GPA|SAT], \\ &= 0.70 + 0.002E[SAT] && \text{by the linearity of expectation,} \\ &= 0.70 + 0.002(1100) = 2.9 && \text{since } E[SAT] = 1100. \end{aligned}$$

- We have that $E[GPA|SAT = 1100] = 0.70 + 0.002(1100) = 2.9$. However, when a student has a $SAT = 1100$ he or she is only expected to have on average a *GPA* of 2.9. However, this only is true *on average*. A particular student with this *SAT* can have a different *GPA* since this is influenced by possibly other variables.